

Predicting Blood Pressure Control Failure and Optimizing Drug Selection in a Hypertensive Patient Population

An Analytical Report Addressing BP Control Prediction and Antihypertensive Drug Effectiveness by Comorbidity Profile

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Executive Summary

Out of 350 patients treated at this hypertension centre, only 33 had their blood pressure under control after three months. That is fewer than 1 in every 10 patients. For the remaining 317, their blood pressure was still dangerously high at the 3month mark, putting them at ongoing risk of stroke, heart attack, and kidney damage.

This report looks at why that is happening and what can be done about it. I analyzed patient records covering age, existing health conditions, prescribed medications, and blood pressure readings before and after treatment. Two questions guided the work. Which patients are most likely to fail, and can we spot them early enough to help? And which medications work best depending on what other health conditions a patient has?

Here is what I found.

The single biggest predictor of failure was how high a patient's blood pressure was when they first came in. The average blood pressure when patients first came in was 166/105 mmHg, which was well above the control target of 140/90 mmHg. Patients who started even higher than this average were the hardest to bring under control by three months. This alone was the strongest signal in the entire analysis.

The second biggest factor was whether patients were actually taking their medication. Patients who took their medication poorly had a control rate of less than 2%. Patients who took it consistently had a control rate of 15%. This difference is significant; however, this is something a doctor can only observe after the first or second visit, not on day one unless the patient's history of adherence is known, which the data provided did not have such information. The practical recommendation here is to check in with patients early, identify those who are struggling with their medication, and act before the three-month review rather than waiting for it.

When it came to which medication worked best, one pattern stood out clearly. Patients on combination therapy, meaning two or more drugs used together, saw an average blood pressure drop of around 22 mmHg, nearly double the drop seen with most single drug treatments. For context, patients on single drug treatments dropped by an average of around 12 to 14 mmHg, not enough to reach the 140/90 target for most patients in this dataset. Their outcomes were better, consistently across almost every patient group. This was true for patients with diabetes, abnormal cholesterol levels, and obesity.

The one exception was patients with chronic kidney disease. For this group, thiazide diuretics performed best with a control rate of 17.6%. This is worth noting separately because kidney disease changes how the body handles fluid and salt, which affects which drug works best.

One finding that deserves attention is the performance of ACE inhibitors. Across every patient group, not a single patient on ACE inhibitors achieved blood pressure control. Interestingly, this drug did produce a meaningful drop in blood pressure readings, just not enough to cross the control threshold. This suggests ACE inhibitors may be too weak as a standalone treatment for patients with blood pressure levels as high as those seen in this dataset.

Based on everything the data showed, three actions stand out for the clinic. Patients who come in with very high blood pressure should be considered for combination therapy from the start rather than trying a single drug first. Patients whose medication history is unknown at their first visit should be followed up within the first few weeks, so that any adherence issues can be caught and addressed before the three month review. And the use of ACE inhibitors as a standalone treatment should be reviewed given its consistent underperformance across all patient groups.

Introduction

High blood pressure, or hypertension, is one of the most common long term health conditions in the world. It rarely causes symptoms on its own, but left uncontrolled it quietly damages the heart, kidneys, and blood vessels over time. The consequences: stroke, heart attack, heart failure, and kidney disease, are among the leading causes of death globally. The good news is that hypertension is treatable. With the right medication and consistent management, blood pressure can be brought under control and these risks can be reduced significantly.

The challenge is that treatment does not always work as expected. Different patients respond differently to medications depending on their existing health conditions, how consistently they take their prescribed drugs, and how high their blood pressure is when they first start treatment. What works well for one patient may do very little for another. This makes hypertension management as much an art of matching the right treatment to the right patient as it is a matter of prescribing the right drug.

This report is built on data from 350 patients treated at a hypertension cardiology centre. Each patient record includes information available at the time of prescription like age, gender, existing health conditions, and the medication they were given, as well as a record of their blood pressure at a three month follow up and whether it was successfully controlled by that point.

The analysis addresses two specific questions. The first is about risk, which patients are most likely to fail blood pressure control at three months, and can we identify them early enough to intervene? The second is about treatment, which medication class produces the best outcomes for patients with specific health conditions such as diabetes, chronic kidney disease, or obesity?

The findings in this report are intended to support better prescribing decisions and earlier clinical intervention. The goal is not to replace clinical judgment but to give doctors and clinic managers a clearer picture of where the risks are highest and where treatment choices can make the most difference.

Data Overview

The dataset for this analysis contains records for 350 patients treated at a hypertension cardiology centre. Each record represents one patient and gives information collected at two different time; when the patient first came in and registered, and again at their three month follow up visit.

The data covers sixteen pieces of information per patient. On the patient side, we have their age, gender, and whether they have any of four existing health conditions: diabetes, chronic kidney disease, abnormal cholesterol levels, and obesity. On the treatment side, it captures which class of blood pressure medication they were prescribed and how consistently they took it. On the clinical outcome side, it records their blood pressure reading before treatment began, their blood pressure reading at the three month follow up, and whether their blood pressure was considered under control at that point.

A patient was considered to have achieved blood pressure control if their reading at the three month follow up was below 140/90 mmHg, the standard clinical threshold for controlled hypertension.

One thing to note about the data is that it was complete. There were no missing values across any of the 350 patient records. This is not always the case with health data and it means the findings in this report are based on a full and uninterrupted picture of every patient in the dataset.

However, there is one important structural issue in the data that shaped how this analysis was done. Out of the 350 patients, only 33 had their blood pressure under control at three months. The remaining 317 did not. This means the data is heavily skewed toward one outcome, failure. In practical terms, this creates a challenge for any predictive model because the model could simply predict failure for every patient and be right nine times out of ten, without actually learning anything useful. To address this, the analysis was designed specifically to give equal attention to both outcomes, ensuring the model learned to identify the minority group, patients who achieved control, just as carefully as the majority.

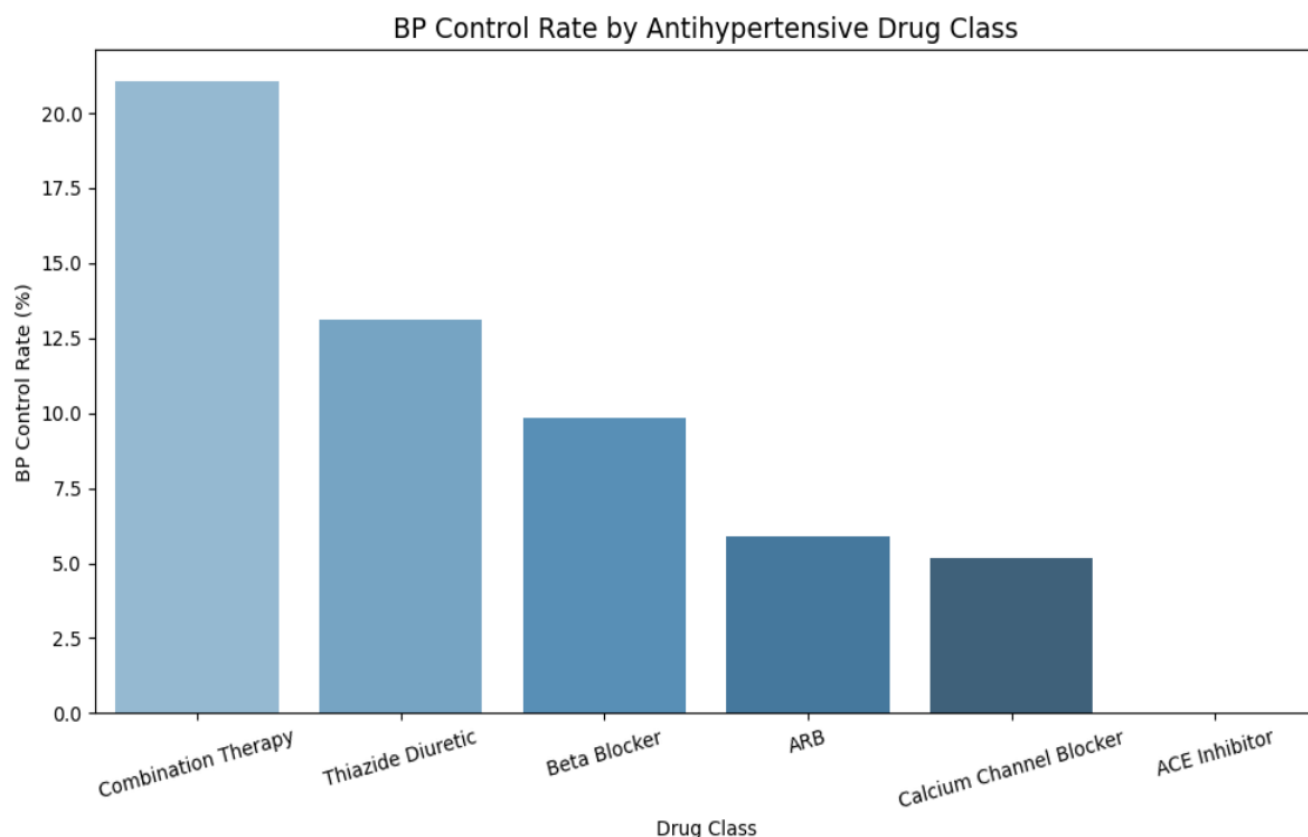
It is also worth keeping something in mind as you read through the findings. When we break those 33 controlled patients into smaller groups, for example, by drug type or by health condition, some groups end up with very few patients in them. A group of 5 or 6 patients is not enough to draw firm conclusions from. If just one or two of those patients had a different outcome, the percentage would look very different. This does not make the findings wrong, it just means they should be read as early signals rather than definitive proof. Wherever this is a concern in the analysis, it is pointed out clearly.

One last thing about the data. 350 patients is a reasonable number to spot patterns and understand what is happening in a clinic. But it is not enough to rewrite treatment guidelines or make complete changes to how patients are managed. Think of the findings in this report as a clear signpost, they point strongly in a direction and they are worth taking seriously. But before any major clinical decisions are made based on them, they should be tested on a much larger group of patients to make sure the patterns hold.

Exploratory Data Analysis

Before building any model, we spent time looking at the data through charts and graphs. The goal was simple, understand what is happening with these patients before drawing any conclusions. Here is what the data showed.

1 Which medication had the best results?

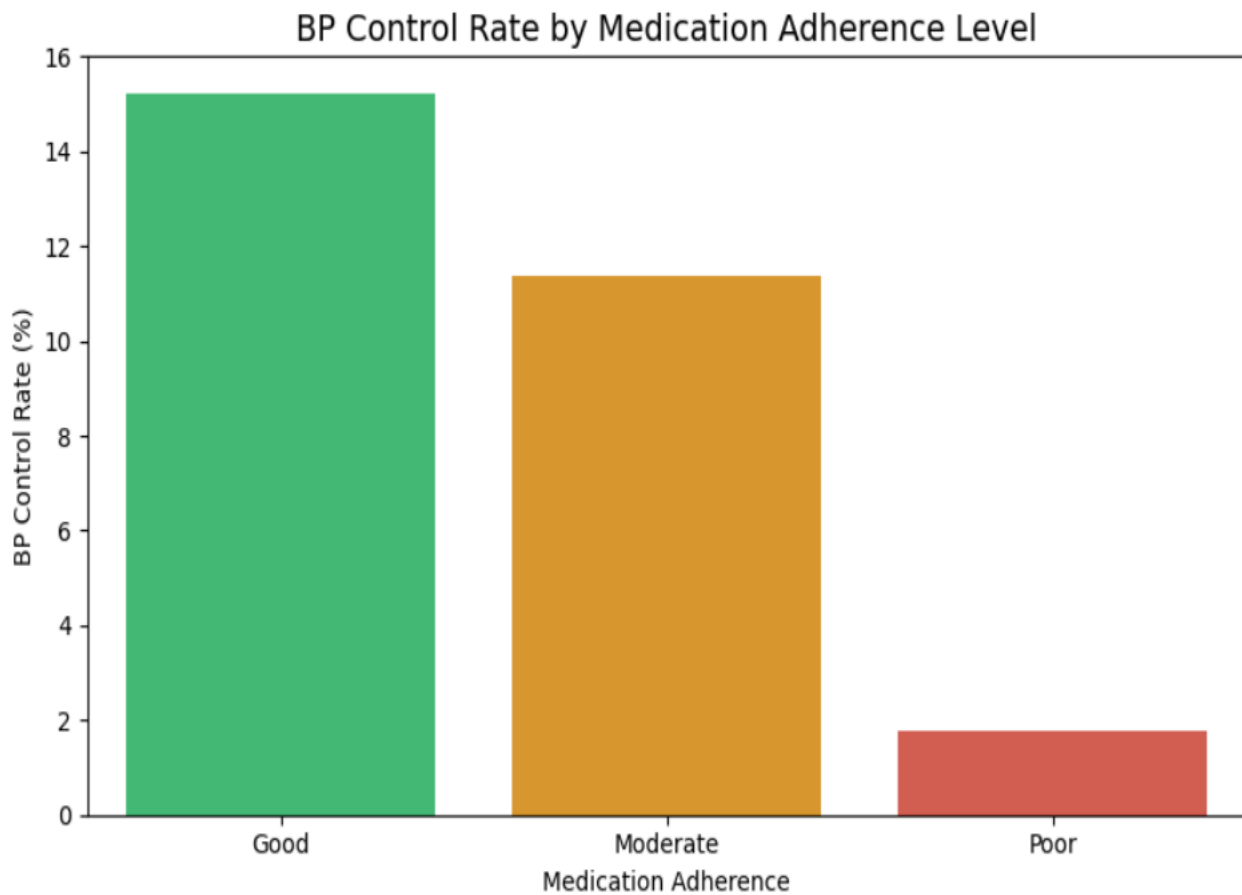


This chart looks at how many patients had their blood pressure under control at three months, grouped by the type of medication they were on.

Combination therapy came out on top. About 1 in every 5 patients on combination therapy had their blood pressure under control, better than any other medication. At the other end, not a single patient on ACE inhibitors achieved control, across 45 patients. That is a result worth paying attention to.

The data point is clearly in one direction, for patients with significantly high blood pressure, a single drug is rarely enough. Combination therapy outperformed every other option and should be the starting point of that conversation at the clinic.

2 Does taking your medication consistently make a difference?

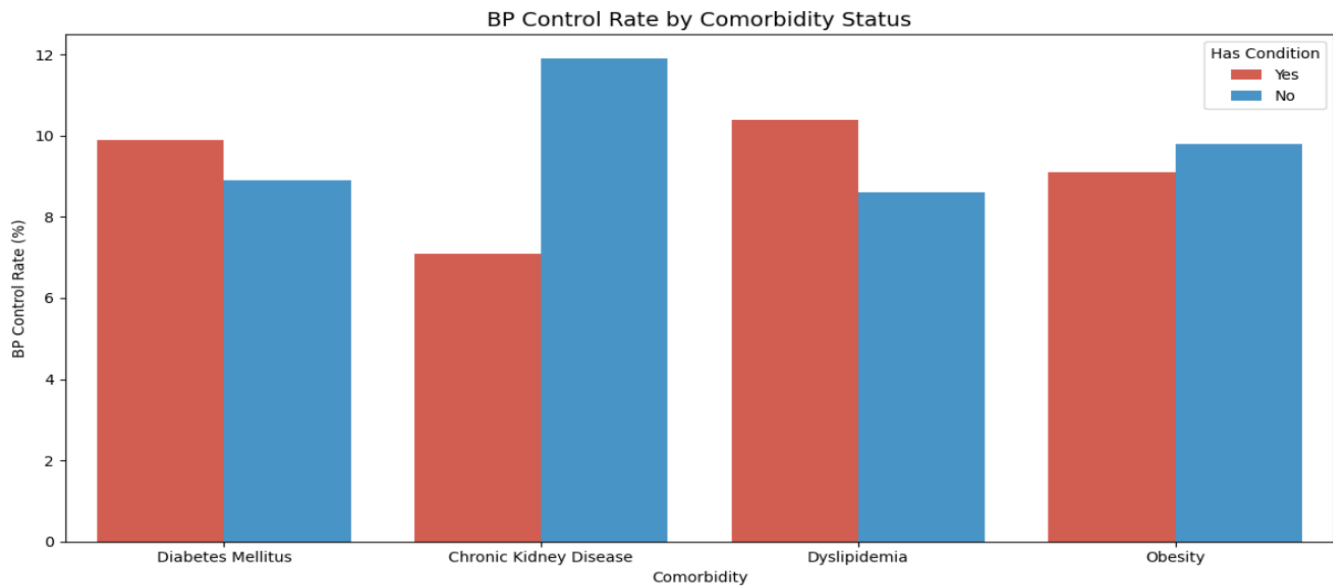


This chart looks at whether patients who took their medication properly had better outcomes than those who did not.

The answer is yes, and the difference is dramatic. Patients who took their medication consistently had a 15% control rate. Patients who took it poorly had less than 2%. Put simply, a medication cannot work if the patient is not taking it. This finding suggests the clinic could improve outcomes significantly just by helping more patients stay on track with their treatment.

Drugs can't work without patients taking them, supporting patients to stay on track with their medication may do more to improve outcomes at this clinic than changing the medication itself.

3 Do existing health conditions affect the outcome?

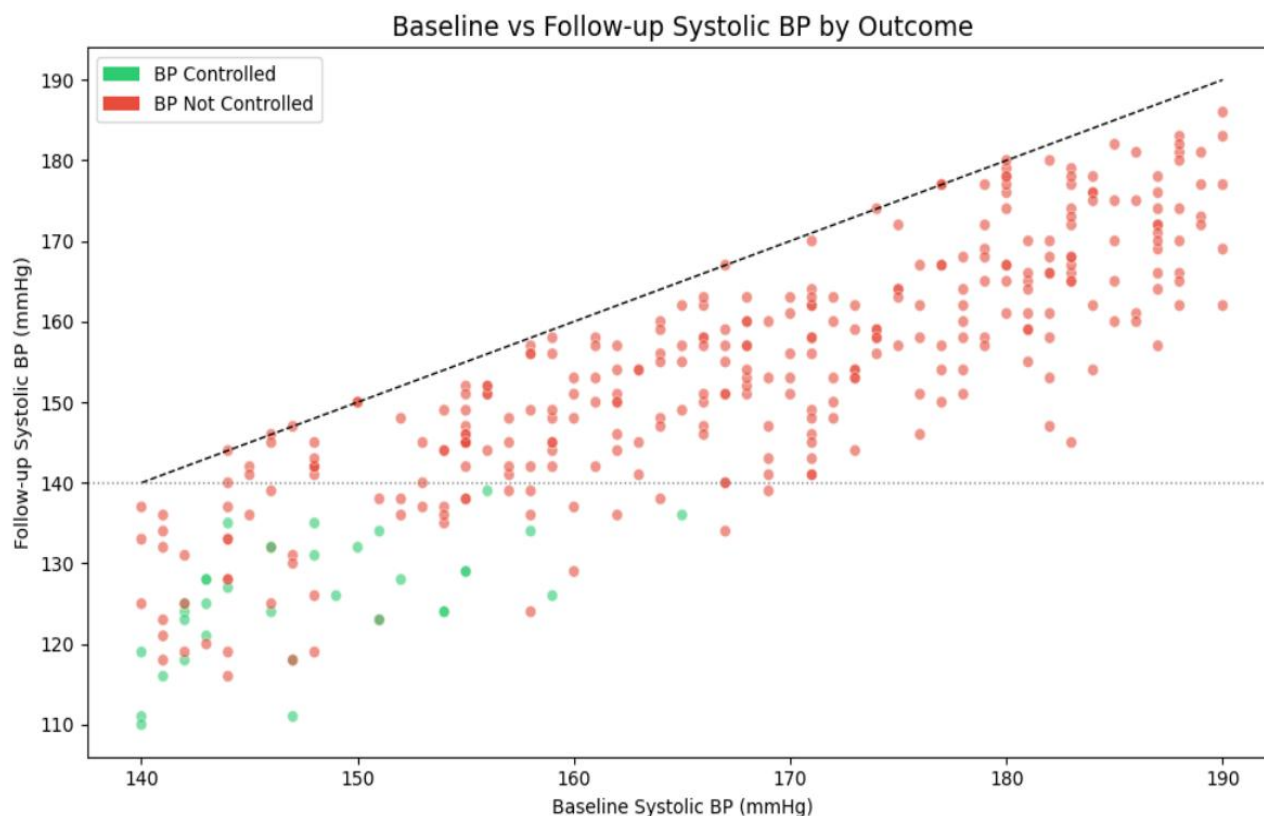


This chart compares outcomes between patients who had each of the four recorded health conditions and those who did not.

Chronic kidney disease made the biggest difference. Patients with kidney disease were nearly half as likely to achieve blood pressure control compared to patients without it. Diabetes, abnormal cholesterol, and obesity did not show a meaningful difference in this dataset. That does not mean those conditions do not matter in general, it just means they did not strongly separate outcomes in this particular group of patients.

Chronic kidney disease stands out as the one condition that meaningfully reduces a patient's chances of getting their blood pressure under control. Patients with kidney disease need closer attention and a more carefully chosen treatment plan.

4 How much did blood pressure actually drop?

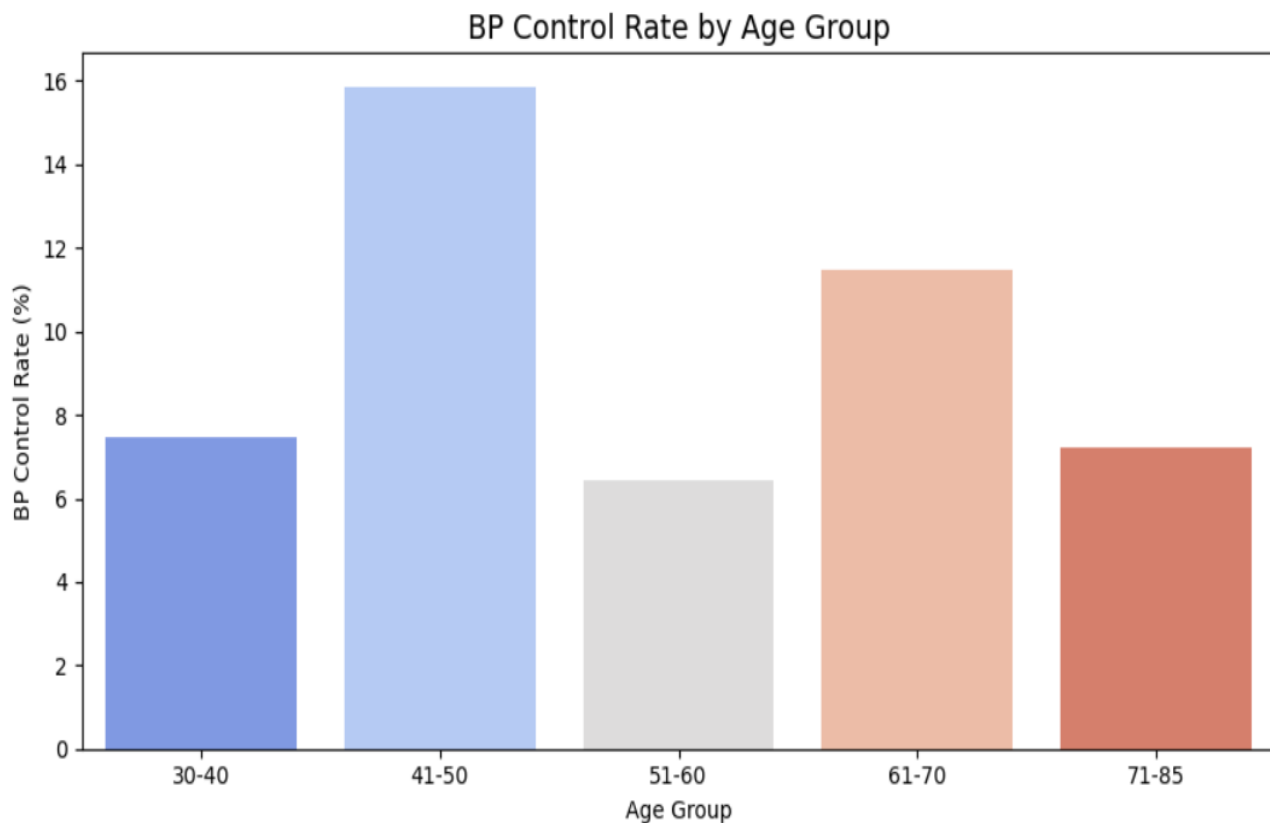


This chart plots where each patient started versus where they ended up at three months, and whether they achieved control or not.

Patients who achieved control saw their blood pressure drop by about 22 mmHg on average. Patients who did not achieve control dropped by about 14 mmHg. That 8-mmHg difference between the two groups is significant. A patient whose blood pressure is barely moving in the first few weeks of treatment is already showing a warning sign. Catching that early, rather than waiting for the three-month review, gives the clinic a real chance to step in and change the outcome.

A blood pressure drop of less than 15 mmHg in the early weeks of treatment is an early warning sign. Patients showing this pattern should be reviewed before the three-month checkpoint, not after it.

5 Does age play a role?

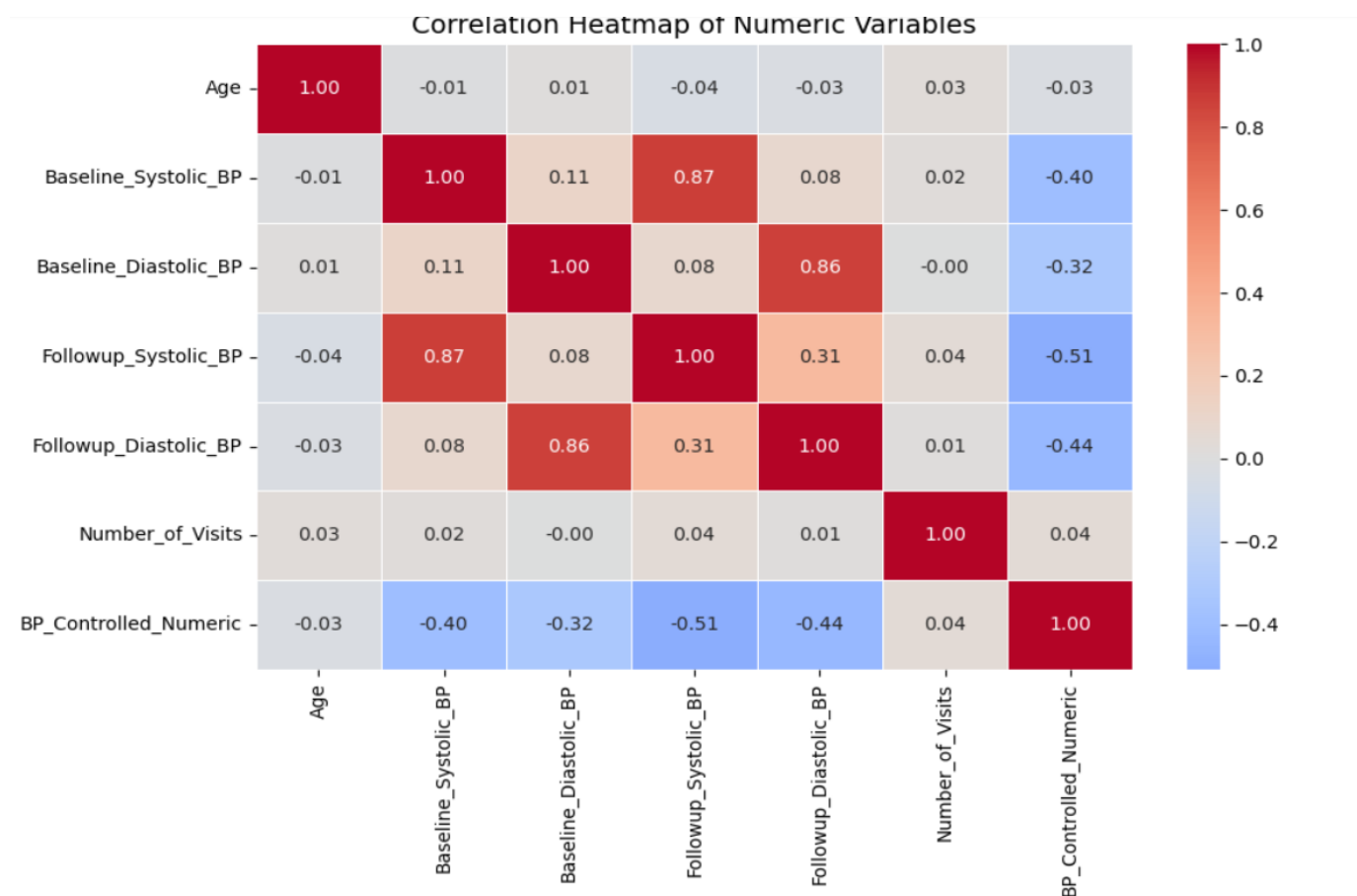


This chart breaks patients into age groups and looks at the control rate within each group.

No clear pattern emerged. Younger patients did not consistently do better than older ones, and older patients did not consistently do worse. The results jumped up and down across age groups without any clear direction. This tells us that age on its own is not a good way to predict who will struggle with blood pressure control. Other things matter far more.

Age is not a reliable guide to who will struggle with blood pressure control. The clinic should not assume younger patients are lower risk or that older patients will automatically be harder to treat.

6 What numbers are most linked to the outcome?



This chart looks at how strongly each measurable variable in the dataset is connected to whether a patient achieved blood pressure control or not.

The clearest finding here was that patients who came in with very high blood pressure were the least likely to achieve control by three months. Both the top and bottom blood pressure numbers at the first visit were strongly linked to the outcome. Age and number of clinic visits showed almost no connection to the outcome at all. This confirms that where a patient starts, their blood pressure on day one, matters more than how old they are or how many times they came in.

Where a patient starts matters more than almost anything else. Patients with higher blood pressure on day one, have a harder road ahead, so should be treated with greater urgency from the very first visit.

Problem 1 — Identifying Patients at Risk of Blood Pressure Control Failure

The Approach

To answer this question, I built a predictive model, that looks at information provided at the provided by a patient at their first visit or shortly after the first visit, and estimate how likely they are to still have uncontrolled blood pressure at three months.

The model was trained on the records of 280 patients and tested on 70 patients it had never seen before. Only information a doctor would realistically have at or shortly after the point of prescription was used: age, gender, existing health conditions, the medication prescribed, how consistently the patient was taking it, and their blood pressure reading on day one. Follow up blood pressure readings were deliberately excluded because a doctor cannot know those at the time of prescribing.

I built and compared two models. The first was a logistic regression model, that not only makes predictions but also explains which factors drive those predictions. The second was a random forest model, a more complex model that is generally more powerful but harder to explain. Both were specifically adjusted to pay extra attention to the smaller group of patients who achieved control, since they made up less than 10% of the dataset.

Choosing the Right Measure of Success

Before looking at results, it is important to explain how I measured whether the model was doing a good job, because the obvious measure turns out to be the wrong one.

The obvious measure is accuracy, how often the model got the answer right. But accuracy is misleading here. A model that predicts failure for every single patient would be right 90% of the time, simply because 90% of patients did fail. That model would never flag a single at-risk patient for early intervention. It would be statistically accurate and clinically useless.

The measure that actually matters here is recall. Recall answers a different question, of all the patients who genuinely needed intervention, how many did the model successfully identify? In a clinical setting, missing a high-risk patient is far more dangerous than raising an occasional false alarm. A missed patient goes without support and may suffer serious consequences by the time the three-month review arrives. A patient who was flagged unnecessarily simply receives a little extra attention, which causes no harm.

For this reason, the model was deliberately set up to prioritize catching as many at risk patients as possible. This is a conscious clinical decision, not a limitation of the model.

How Well Did the Model Perform?

The logistic regression model correctly identified 6 out of every 7 patients who achieved blood pressure control in the test group. The random forest model only identified 3 out of 7, missing more than half the at-risk cases.

Overall, the logistic regression model was correct 98% of the time. The random forest scored marginally higher on this measure but its poor performance on recall made it the wrong choice for a clinical tool where missing patients carries real consequences.

Logistic regression was therefore selected as the primary model, not because it scored highest on every metric, but because it performed best on the metric that matters most in this setting.

Threshold Tuning

By default, the model flags a patient as high risk when it calculates a 50% or higher probability of control failure. I tested whether adjusting this threshold could improve performance further.

At the default threshold of 0.5, the model missed one high risk patient out of seven. By lowering the threshold to 0.3, the model successfully identified all seven controlled patients in the test group without missing a single one. The cost of this adjustment was eight false alarms, meaning eight patients received additional attention they did not strictly need.

In a clinic where missing a high-risk patient can lead to stroke, heart attack, or kidney failure, eight unnecessary follow up calls is a small and manageable price to pay. A threshold of 0.3 is therefore recommended as the operating point for this model.

Threshold	Patients Caught	Patients Missed	False Alarms	Recall
0.5 — default	6 of 7	1	5	86%
0.3 — recommended	7 of 7	0	8	100%

What Predicts Failure — The Risk Factors

When the model was asked which factors most strongly predicted whether a patient would fail blood pressure control, three stood out clearly above everything else.

The strongest predictor was how high a patient's blood pressure was when they first came in. Patients presenting with very high baseline readings were significantly less likely to reach the control target by three months. This was the single most powerful signal in the entire model, more powerful than any health condition, any medication, or any demographic factor.

The second strongest predictor was medication adherence. Patients observed to be taking their medication poorly in the early weeks of treatment were far less likely to achieve control. It is important to be clear about one thing here; adherence cannot be measured on day one. It only becomes visible after at least one follow up visit. This means the model works best not at the very first appointment but at the early review, where adherence can be assessed and acted on before the three-month deadline.

Drug class was the third meaningful factor. The choice of medication was associated with noticeably different outcomes across patient groups, reinforcing that treatment selection matters and is explored in detail in Problem 2 of this report.

Gender, diabetes, and chronic kidney disease made smaller but still notable contributions. Age, obesity, and abnormal cholesterol showed very little predictive power on their own in this dataset.

Actions arising from these findings, including guidance on early intervention and follow-up timing, are outlined in the Recommendations section.

Problem 2 — Which Medication Works Best for Each Type of Patient?

The Approach

Not all hypertensive patients are the same. A patient who also has diabetes faces different biological challenges than a patient who also has chronic kidney disease. The medication that works well for one may do very little, or even cause harm, for the other.

To answer this question, I grouped patients by their existing health conditions and looked at how each medication performed within each group. Two things were measured for every drug and condition combination. The first was the blood pressure control rate, the percentage of patients in that group whose blood pressure was successfully brought under control at three months. The second was the average blood pressure drop, how many mmHg the blood pressure fell on average, regardless of whether it crossed the control threshold.

Looking at both measures together gives a holistic picture than either one alone. A drug might produce a large blood pressure drop but still not bring patients below the control threshold. Another drug might have a modest drop but consistently get patients over the line. Both pieces of information matter when making prescribing decisions.

It is important to note that patients in this dataset were not randomly assigned to medications. Doctors made prescribing decisions based on their own clinical judgment. This means we cannot say with certainty that the medication caused the outcome, only that certain medications were associated with better outcomes in certain patient groups. With a larger dataset, stronger conclusions could be drawn.

The Overall Picture

Before looking at individual conditions, one pattern stood out across the entire dataset. Combination therapy, meaning two or more blood pressure medications used together, consistently outperformed every single drug used alone. This was true not just overall but within almost every patient subgroup examined.

This finding makes clinical sense. The patients in this dataset came in with an average blood pressure of 166/105 mmHg, well above the control target of 140/90 mmHg. For many of these patients, one drug simply does not have enough power to bridge that gap on its own. Using two drugs together, each working through a different mechanism and pathway, gives the treatment a much better chance of reaching the target.

It is worth noting that the control threshold used throughout this analysis is 140/90 mmHg. A drug that reduces blood pressure meaningfully but leaves a patient above this threshold is still recorded as a failure, which explains why some medications show a reasonable average blood pressure drop but a low control rate.

What Works for Each Patient Group

1. Patients with Diabetes

For patients with both hypertension and diabetes, combination therapy delivered the best results by a clear margin. It achieved a control rate of 25% and an average blood pressure drop of 22.7 mmHg, the highest of any drug in this group. Every other medication fell well below this, with control rates ranging from 3.8% to 11.1%. For diabetic patients whose blood pressure is significantly elevated, starting with a single drug appears unlikely to be sufficient based on this data.

2. Patients with Chronic Kidney Disease

This group told a different story, and it is the most important exception in the entire analysis. For patients with chronic kidney disease, thiazide diuretics performed best with a control rate of 17.6%. Combination therapy came second at 11.5%. This is worth paying close attention to. Kidney disease changes how the body handles fluid and salt, and thiazide diuretics work partly by helping the kidneys clear excess fluid, which directly addresses one of the mechanisms driving high blood pressure in this patient group.

ACE inhibitors recorded a 0% control rate in CKD patients despite producing an average blood pressure drop of 15.7 mmHg. This suggests they are reducing blood pressure but not enough to cross the control threshold. More importantly, some ACE inhibitors are known to place additional strain on already damaged kidneys. The zero percent control rate in this group warrants a careful clinical review of how ACE inhibitors are currently being used with kidney disease patients at this clinic.

3. Patients with Abnormal Cholesterol

Combination therapy again led this group with a control rate of 25.9% and an average blood pressure drop of 22.5 mmHg. Thiazide diuretics came second at 16.7%. ACE inhibitors once again recorded zero successful outcomes despite producing a measurable blood pressure reduction of 15.9 mmHg on average.

4. Patients with Obesity

For obese patients, combination therapy led with a control rate of 18.4% and an average drop of 21.6 mmHg. ARBs and thiazide diuretics performed reasonably in second and third place. ACE inhibitors again recorded no successful outcomes in this group.

The ACE Inhibitor Finding

Across every single patient group in this dataset; diabetes, chronic kidney disease, abnormal cholesterol, and obesity, not one patient on ACE inhibitors achieved blood pressure control at three months. This is consistent and impossible to ignore.

However, two things must be said alongside this finding. First, ACE inhibitors did produce a meaningful blood pressure drop in every group, averaging between 14 and 16 mmHg. The problem is not that they do nothing. The problem is that they do not do enough to get patients across the finish line, particularly when starting blood pressure levels are as high as those seen in this dataset. Second, ACE inhibitors are used across 45 patients in total, a small number. The finding is consistent but the dataset is not large enough to make this a definitive clinical verdict. What it is large enough to do is raise a serious question that deserves investigation with more data.

Summary Table

Patient Profile	Best Drug Class	Control Rate	Avg BP Drop	Second Best
Hypertension + Diabetes	Combination Therapy	25.0%	22.7 mmHg	Thiazide Diuretic
Hypertension + CKD	Thiazide Diuretic	17.6%	12.9 mmHg	Combination Therapy
Hypertension + Abnormal Cholesterol	Combination Therapy	25.9%	22.5 mmHg	Thiazide Diuretic
Hypertension + Obesity	Combination Therapy	18.4%	21.6 mmHg	ARB

Prescribing recommendations based on these findings are outlined in the Recommendations section.

Conclusion

Out of 350 patients treated at this centre, only 33 had their blood pressure under control after three months. That means for every 10 patients who walked in and started treatment, 9 of them left that three-month review still at risk. That is the problem this report set out to understand.

What the data showed is that two things drive this more than anything else. First, how high a patient's blood pressure is when they first come in. Patients who start very high have a much longer distance to travel to reach a safe level, and one medication is often not strong enough to get them there. Second, whether the patient is actually taking their medication. Patients who were not taking their medication consistently had almost no chance of getting their blood pressure under control, less than 2 in every 100. No medication in the world works if it is not being taken.

On the question of which medication works best, the picture was fairly clear. For most patients, including those with diabetes, abnormal cholesterol, and obesity, using two medications together worked better than using one alone. The only group that was different was patients with kidney disease, where a drug called thiazide diuretic gave the best results. One drug, ACE inhibitors, did not produce a single successful outcome across any patient group in this dataset. That finding alone deserves a closer look from the clinical team.

The path forward for this clinic does not have to be complicated. Patients who come in with very high blood pressure should be started on two medications from the beginning rather than trying one and waiting three months to see if it works. And every patient should be contacted early, within the first few weeks, to make sure they are taking their medication and that it is beginning to have an effect. By the time the three-month review arrives, the clinic should already know whether a patient is on track. That review should be a confirmation, not a surprise.

Limitations

Every analysis has boundaries; things the data cannot tell us. Being clear about those boundaries is not a weakness. It is what makes the findings trustworthy.

The dataset is small. 350 patients are enough to spot clear patterns and point in a meaningful direction. But it is not enough to rewrite clinical guidelines. Some of the subgroups in this analysis, particularly when patients were split by both health condition and medication type, contained as few as 20 to 30 patients. A single patient changing outcome in a group that small can shift the percentage significantly. The findings here should be treated as strong early signals that deserve further investigation with a larger patient population before any formal protocol changes are made.

Patients were not randomly assigned to medications. Doctors chose which medication to prescribe based on their own clinical judgment. This means patients with more severe conditions may have been steered toward certain drugs, and patients with milder presentations toward others. This makes it difficult to say the medication caused the outcome, only that certain medications were associated with better outcomes in certain groups. A properly controlled study would be needed to establish cause and effect with confidence.

Medication adherence could not be measured at day one. Adherence was recorded in the dataset but it reflects behaviors observed over time, not something a doctor can assess at the very first appointment with a new patient. This limits the model's ability to be a true point of prescription tool for patients with no prior history at the clinic. It works best as an early intervention tool, most useful at the first follow up visit rather than on day one.

The reason for ACE inhibitor underperformance is not fully explained by this data. ACE inhibitors recorded zero successful outcomes across all patient groups. The data shows this clearly but cannot fully explain why. It could be related to the severity of patients assigned to this drug, dosing practices, or other factors not captured in the dataset. Further investigation is needed before drawing firm clinical conclusions about this drug class.

The dataset covers one centre only. These findings reflect the patient population, prescribing practices, and clinical environment of one specific hypertension centre. The patterns may look different at another centre with a different patient mix. The recommendations in this report should be applied with that context in mind.

Recommendations

Based on the findings of this analysis, the following actions are recommended for the clinic. These are listed in order of priority.

1. Start high risk patients on combination therapy from day one: Patients who come in with a systolic blood pressure significantly above 166 mmHg, the average seen in this dataset, should be considered for combination therapy immediately rather than starting with a single drug and waiting three months to see if it works. The data shows that single drug treatments rarely achieve control in this patient population, and three months is a long time to leave a high-risk patient on an ineffective treatment.

2. Schedule every patient for an early follow up within two to four weeks: The three-month review should not be the first time the clinic checks in after prescribing. An early visit within the first two to four weeks serves two purposes, checking whether the blood pressure is beginning to respond, and finding out whether the patient is actually taking their medication. Catching problems at this stage leaves enough time to act before the three-month deadline.

3. Prioritise adherence support for patients showing early signs of poor medication taking: Patients observed to be taking their medication poorly in the early weeks had a control rate of less than 2%. Simple interventions, reminder calls, simplified dosing instructions, or a brief conversation about why the medication matters, could move patients from poor to moderate adherence and significantly improve their chances of control.

4. Review the use of ACE inhibitors as a standalone treatment: ACE inhibitors recorded zero successful outcomes across all patient groups in this dataset. While the sample size limits a definitive verdict, the consistency of this finding across 45 patients and four different health condition groups is strong enough to warrant a clinical review. Combining ACE inhibitors with another agent or replacing them with combination therapy for significantly hypertensive patients may produce better outcomes.

5. Apply different treatment approaches for patients with chronic kidney disease: CKD patients responded differently to medication than every other patient group. For this group specifically, thiazide diuretics outperformed combination therapy. Prescribing decisions for CKD patients should reflect this distinction rather than applying a one size fits all approach.

6. Use the predictive model as an early intervention tool at the first follow up visit: The model built in this analysis can identify high risk patients with a recall of 100% at a threshold of 0.3. This means it can flag every patient likely to fail control before the three-month review arrives. The recommended workflow is to run the model at the first follow up visit, once adherence can be observed, and use the output to prioritise which patients need more intensive support in the weeks that follow.